Lorem ipsum

Running title: Hipster ipsum is better

Patrick D. Schloss †

†To whom correspondence should be addressed:

pschloss@umich.edu

Department of Microbiology & Immunology

University of Michigan

Ann Arbor, MI 48109

Research article

Abstract

Sus umami skateboard activated charcoal XOXO cornhole, thundercats pitchfork. Succulents street art semiotics, artisan lo-fi same food truck forage aesthetic plaid drinking vinegar ugh slow-carb. Sustainable vexillologist ramps cold-pressed, selfies pop-up microdosing messenger bag slow-carb. Lo-fi fingerstache cred, copper mug chicharrones DSA chambray. Craft beer meditation shabby chic, put a bird on it tonx ramps tumeric palo santo etsy dreamcatcher chartreuse snackwave. Fashion axe kogi sus, migas etsy gluten-free semiotics hammock chartreuse cardigan.

Importance

Taiyaki hoodie letterpress, franzen man braid bushwick food truck humblebrag chia palo santo vinyl keffiyeh. IPhone ugh kitsch tousled tumblr messenger bag, kogi actually taiyaki. Austin taiyaki succulents vaporware letterpress bruh. Wayfarers listicle lyft beard tousled keffiyeh fingerstache neutral milk hotel ramps prism. Tote bag vape gentrify, fingerstache fixie marxism enamel pin pork belly cardigan microdosing hoodie. Freegan Brooklyn biodiesel, fanny pack actually kombucha organic whatever literally crucifix. Fingerstache kombucha subway tile squid, marxism kogi VHS ethical distillery fanny pack jean shorts.

Introduction

(1, 2)

A persistent challenge in amplicon and shotgun metagenomic sequencing of microbial communities is the variation in the number of reads obtained across samples [**XXXXX**]. The variation in the number of reads generally occurs because of an accumulation of sampling errors during library generation that includes attempts to pool equimolar amounts of DNA for subsequent sequencing. It is not uncommon to observe as much as 100-fold variation in sequencing depth [**XXXXX**]. This challenge is not unique to microbial ecologists and has long plagued landscape and animal ecologists [**XXXXX**]. The classical approach to control for uneven sampling has been rarefaction [**XXXXX**]. In rarefaction, a consistent number of individuals are randomly sampled from each sample. An alpha or beta diversity metric is then calculated from the sampling. The random subsampling and diversity measurement steps are repeated a large number of times (e.g., 100 or 1,000) and the measurements are averaged over the subsamples [**XXXXX**]. Unfortunately, the term “rarefying” and occasionally rarefaction have been misidentified as a single subsampling step [**XXXXX**]. Regardless, over the past 10 years, the process of rarefaction has become controversial because some view the choice of the sampling depth as arbitrary, the metrics can become more similar than they truly are, and because the practice of downsampling to a common number of sequences leaves the impression that good data are being unnecessarily discarded [**XXXXX**]. Numerous alternatives have been proposed including calculating the relative abundance of each taxon in a community so that they sum to 1 within a sample [**XXXXX**], normalization of counts so that the total number of individuals sums to a user defined value [**XXXXX**], use of estimators of diversity that can extrapolate beyond the number of observed individuals [**XXXXX**], and the use of compositional statistics, which are based on a log ratio of taxon counts to a reference value [**XXXXX**].

Researchers have adopted compositional methods because the number of sequences obtained per sample is typically independent of the true number of individuals in a community and so the

number of times each taxon is observed is not independent of the other taxa [**XXXXX**]. The general approach is to calculate a centralized log ratio (i.e., CLR) where the abundance of each taxon within a sample is divided by the geometric mean abundance within the sample, which is then used as the argument for the natural logarithm. These values are then used to calculate pairwise euclidean distances between samples resulting in so-called Aitchison distances [**XXXXX**]. Aside from controlling for the compositional nature of sequence data, proponents of the Aitchison distance claim that the distances are insensitive to differences in sampling effort [**XXXXX**]. A challenge of Aitchison distances is that because microbial communities are patchy a large number of features will have a frequency of zero in each sample. This will cause the geometric mean to be undefined. To overcome this challenge, factors are added to the data. Three approaches have been proposed: adding a pseudocount of 1 all frequency data (i.e., Nudge CLR), adding a pseudocount of 1 divided by the total number of individuals in the sample (i.e., Nudge CLR), and imputing zero values to be a random small value (i.e. Zero CLR) [**XXXXX**]. As an alternative to these transformations, Martino et al. [**XXXXX**] (i.e., Martino) proposed calculating CLR values for each sample after ignoring zero values followed by matrix completion or robust principal components analysis (i.e., RPCA). The authors claimed that the advantage of RPCA over the other CLR-based approaches was that it did not require imputing zero values and that the method was less sensitive to uneven sampling effort.

In contrast to these claims, I have shown that the output of RPCA and other alternatives to rarefaction are more sensitive to uneven sampling than rarefaction [**XXXXX**]. Furthermore, use of these methods can inflate the false detection rate and decrease the statistical power to detect differences in alpha and beta diversity metrics. Given the near unanimous agreement over the past decade that rarefaction is “inadmissible” and the development of alternatives to rarefaction, one might question why my results disagree so markedly with those of others. There are several possible reasons. First, researchers often propose new methods by demonstrating their method reveals new insights over old methods without objectively benchmarking the method [**XXXXX**]. Because one cannot know the true difference between samples, a positive or negative result with

experimental data cannot be interpreted. Second, when benchmarking tests are developed, researchers may inadvertently select tests that favor the method they are developing. This was the case in my recent re-analysis of the original paper that called rarefaction into question [**XXXXX**]. Therefore, it is necessary to use a comprehensive benchmarking test suite that allows one to compare diverse methods using inference methods that are commonly used by researchers. Third, the choice of how to simulate community data can inadvertently favor the method being developed. A popular cliché states that "all models are wrong, some are useful" [**XXXXX**]. Similar to one's choice of benchmarking tests, the modelling approach may favor some methods over others. For instance, I have critiqued how others have accounted for variation in sampling depth as being unlike what we see in experimental data [**XXXXX**].

In my previous analyses advocating for rarefaction, I generated null community models and treatment groups with known effect sizes by creating taxon abundance distributions and sampling depths based on 12 published datasets [**XXXXX**]. Alternatively, Martino modelled communities with known effect sizes using a statistical model that sought to mimic the diversity of communities described in a study comparing the microbial communities found on people's fingertips and their computer keyboards [**XXXXX**]. As described in Martino, the overall simulation framework involved simulating two treatment groups each represented by 100 communities for a total of 200 communities. Each community contained as many as 1,000 taxa, which they referred to as features. For each treatment group they fit a normal distribution to the relative abundances of taxa from two individuals' samples to obtain a mean and standard deviation for each taxon. The fraction of taxa that overlapped between treatment groups was also selected to mimic the overlap observed between the two individuals. The expected taxon relative abundances for each sample were the same across each sample within the same treatment group. Next, a common normal distribution with a single mean and standard deviation was used to generate noise (i.e., homoscedastic noise) that added to each relative abundance value to add variation across samples and features. Next, a random subset of sample-taxa combinations were selected for incorporation of additional noise (i.e., heteroscedastic noise). Finally, the sample-taxa

combinations were randomly sampled using a Poisson-lognormal distribution to generate the simulated number of sequences for each sample-taxa combination so that the total number of sequences within a sample contained approximately 1,000, 2,000, 4,000, or 10,000 sequences per simulation. One set of 200 samples were simulated for each sequencing depth. In addition to the simulation where a subset of taxa was shared in both treatment groups, Martino created positive and negative control simulations. In the positive control simulation, there were no overlapping taxa resulting in two clearly defined groups of samples. In the negative control simulation all the taxa were shared in similar abundance so that the groups of samples could not be distinguished. Finally, they compared RPCA to other methods using two published datasets.

This difference in modelling strategy has created an interesting opportunity to reframe whether to use rarefaction and to reassess the performance of RPCA and other CLR-based approaches. Although Martino used an interesting modelling approach, they did not compare RPCA to other CLR-based approaches or to rarefaction. Furthermore, they used benchmarking tests that did not reflect how researchers typically analyze microbial community data. Therefore, the goal of the current study was to benchmark RPCA and other CLR-based methods relative to rarefaction using the modelling framework used by Martino.

Results

Description of simulated communities

Martino made a git-based repository available on GitHub that includes the actual benchmarking code written in Python and the data used in their analysis (https://github.com/cameronmartino/deicode-benchmarking); the code and data from that repository are included in the repository for the current analysis (https://github.com/SchlossLab/Schloss_RPCA_XXXXXX_2024). To maintain a connection to the original simulations and implement a broader set of tests, several changes were necessary in the implementation of Martino's code. First, because one goal of this study was to assess the impact of variation in sequencing effort on the performance of RPCA, I modified the parameters inputted

to Martino's code to produce more than the desired number of sequences per sample and then performed a single subsampling of the samples to 1,000, 2,000, 4,000, or 10,000 sequences per sample to have exactly the desired number of sequences per sample. Second, to account for the effect of the random number generator, instead of replicating each simulation condition only once as Martino did, I replicated each condition 10 or 100 times depending on the analysis. Third, Martino only considered the positive and negative controls with 4,000 sequences, my implementation of their simulations and subsequent tests incorporated the sequencing depths of 1,000, 2,000, and 10,000 that were used in the simulations with partially overlapping treatment groups. The regenerated heatmaps and principal coordinates analysis (PCoA) plots in **Figure 1** are qualitatively similar to those in Figure 3 of Martino. The most notable difference with Martino is that they inadvertently switched the labels of the axes and performed the OCLR transformation on features rather than samples (**Supplementary Text**). Replicating what Martino reported, the first two axes of the Bray-Curtis ordinations only explain a fraction of the total variation in the data while the RPCA ordinations explained all the variation. This was because RPCA is run by giving the algorithm a desired number of axes; in this case, 2 axes. The ratio of the fraction of the variation explained by the first two axes were similar between RPCA and PCoA with Bray-Curtis.

Comparison of simulated communities using PERMANOVA

Martino used the PERMANOVA F-statistic to compare the ability of RPCA to differentiate treatment groups when using raw counts and when removing zero values prior to calculating the CLR (RCLR). Using the partially overlapping treatment groups they found that the RCLR pretreatment yielded larger F-statistics than using raw counts. Repeating Martino's PERMANOVA analysis, I replicated their results when using RPCA (**Figure 2**). However, by generating 10 replicate datasets (Martino only performed one), I was observed considerable variation across replicates for the F-statistics calculated using RPCA with raw counts. Unfortunately, Martino did not make any comparisons to other methods of calculating distances. Calculating Euclidean distance using other central log ratio and DeSeq2 transformations and Bray-Curtis distances

using raw counts and CSS and SRS normalized data all resulted in small F-statistics. I suspected that the RPCA data generated relatively large F-statistics because the Euclidean distances calculated using RPCA was only done with the first two axes. This would have removed a considerable amount of variation that was embedded in the third and higher axes. The result would be elevated F-statistics. To test this hypothesis, I performed PCoA on Bray-Curtis distances, calculated Euclidean distances using the first two axes of the resulting PCoA, and subjected those distances to PERMANOVA (see "Bray-Curtis PCoA" data in the Euclidean distance section of **Figure2**). The calculated F-statistics were even larger than those generated by RPCA. This suggested that RPCA may make communities appear more similar than they are when differences are more pronounced in higher axes. Regardless of the relative scale of the F-statistics, the replicates for all approaches yielded significant p-values except for those generated using RPCA with raw counts.

Comparison of simulated communities using KNN Classification

In addition to comparing F-statistics from PERMANOVA, Martino used KNN classification to distinguish between the two partially overlapping treatment groups. Although Martino's text indicated they used 40% of the samples in the training set and 60% in the testing set (i.e., a 40:60 cross validation split), their code indicates it was actually the opposite with 60% of the samples in the training set and the remaining in the testing set (**Supplemental Text**). To execute the KNN classification, Martino used the first two axes calculated from RPCA using raw counts and RCLR transformed data as the input to their KNN classification. Because I was interested in comparing other transformations and the use of Bray-Curtis distances, I wrote a KNN classification procedure that would allow me to input distance matrices. Furthermore, I replicated the 60-40 split 100 times for each of the 10 simulations reporting the median accuracy on the 40% testing data for each simulation. The original Martino data indicated that the accuracy of using raw counts with RPCA was near 0.5, indicating it did no better than randomly guessing. While the minimum values I observed across the 10 replicate simulations overlapped with these

results, the median values were all better than random ("Raw + RPCA" in **Figure 2**). I replicated Martino's result showing that using RCLR transformed counts as input to RPCA yielded higher accuracy than using raw counts.

Next I sought to place the RPCA results in context with other methods. I genenerated Euclidean distance matrices using other central log ratio methods and DeSeq2. I also generated Bray-Curtis distance matrices using the raw counts and CSS and SRS Normalization. These methods all yielded similar results, likely because all of samples in the simulations had the same sequencing depth. Compared to these other approaches, RPCA was more sensitive to differences in sequencing depth. Although accuracy values across all methods were comparable at 4,000 and 10,000 sequences, the RPCA-based methods underperformed the others at 1,000 and 2,000 sequences.

Martino did not report the classification accuracy for their positive and negative controls. As expected, the mean accuracy for classifying the treatment groups in the negative control model across each depth, transformation, and distance calculation was 0.50 (sd = 0.0085). Furthermore, the mean accuracy for classifying samples in the positive control was 0.99 (sd = 0.0086).

Rather than using a training data set to identify an optimal number of neighbors (i.e., K), Martino used 13 neighbors. To assess whether the classification was sensitive to the number of neighbors, I split the 60% of samples in the training data into another 60-40 split to train and optimize the number of neighbors 1,000 times [**XXXXX**]. Across the various conditions, the median optimal number of neighbors was 33.00 (IQR=21.00 to 41.00). When I instead performed 1,000 80-20 splits on the full and training datasets, the median optimal number of neighbors was 58.00 (IQR=42.00 to 65.00). Regardless of finding that the optimal number of neighbors was consistently larger than the 13 that Martino used, the difference in accuracy values was only modestly better than the other fitting procedures when using 80-20 splits (**Figure S1**).

Comparing PERMANOVA and KNN Classification results with more similar treatment groups

Using the simulation where the treatment groups partially overlapped, the PERMANOVA analyses yielded significant P-values and KNN classification were effective at distinguishing between the groups for nearly all of the methods. Because it was not possible to easily differentiate between the methods, I created treatment groups that were more similar than those simulated by Martino. Unfortunately, in the process of modifying their code, I identified a number of incongruities between Martino's written description and their code that limited my ability to reuse their code to simulate greater overlap between the treatment groups (**Supplemental Text**). These issues included: (i) samples rather than features were shared between treatment groups; (ii) among samples from the same treatment group, the relative abundance distributions for the features were uniform; (iii) the total number of individuals in the samples varied across the simulated datasets; (iv) the heteroscedastic source of noise was actually applied to all sample-feature combinations; (v) negative mean values were used when sampling normal distribution sampling functions when calculating the heteroscedastic noise; (vi) negative abundance values generated after applying the heteroscedastic noise were made positive; and (vii) setting the random number generator seed was unable to produce reproducible simulations. I rewrote the simulation code in the R programming language to correct these issues. To simulate a more nuanced difference in treatment groups, I increased the total number of features across all samples from 1,000 to 10,000 and set 12 features to be differentially abundant between the treatment groups; these values were selected to create conditions where approximately 50% of PERMANOVA tests using Bray-Curtis distances yielded P-values less than 0.05. Using the same number of features, I recreated negative and positive control simulations in which all features were shared or where the features were differentially represented between one of the two treatment groups. The resulting simulations provided more realistic benchmarking data than which was observed by Martino (**Figure 3**). The simulations were replicated 100 times to obtain greater precision in calculating the power to detect differences by PERMANOVA and the

accuracy of KNN classification.

Using the revised simulation data revealed a considerable difference in the PERMANOVA and KNN classification relative to the original simulation. Among the various methods, the RPCA-based methods performed the worst by both PERMANOVA and KNN classification (**Figure 4**). There was effectively no statistical power to detect differences using RPCA or to classify samples using KNN classification even at the largest sequencing depths. It was not possible to distinguish the performance between the various CLR and DeSeq2 methods using Euclidean distances or any of the methods when Bray-Curtis distances were calculated. Among these methods, the power to detect differences was routinely above 0.75 when 10,000 sequences were generated per sample. For these methods, there was a clear relationship between the number of sequences obtained from each sample and the observed power and accuracy. Similar to the earlier comparison of approaches for fitting the number of neighbors, the performance of KNN classification was modestly better using 80-20 splits with a grid-based search (**Figure S2**). From the revised simulations, it is evident that the performance of RCLR and RPCA was considerably worse than the other methods.

Description of case study datasets

After describing the results of their simulations, Martino and colleagues used two published datasets as case studies to evaluate RPCA and related methods. They evaluated RPCA relative to Bray-Curtis and weighted Unifrac distances using PERMANOVA, KNN, and ordinations. The two datasets were drawn from a natural study comparing healthy and stressed sponges [**XXXXX**] and from another study of a murine model of sleep apnea were mice were given air or intermittent hypoxia and hypercapnia (IHH) [**XXXXX**]. Both datasets were small. The sponge dataset had 133 healthy and 85 stressed samples and the sleep apnea dataset had 96 air treated and 93 IHH treated samples (**Figure 5**). The effect sizes for both studies were relatively large and could easily be differentiated by the first two PCoA axes. Similar to the results from the simulations, I wondered whether a difference was observed between the methods when there

was a smaller effect size and when there was better control for differences in effect sizes. Therefore, I added a third dataset from a study that explored associations between the microbiome and colorectal cancer [**XXXXX**]. The differences between the two treatment groups were not so obvious by PCoA and there were more subjects. The colorectal cancer dataset included 261 subjects with healthy colons and 229 subjects with screen relevant colonic neoplasias (**Figure 5**). Sequences from the three datasets were processed in parallel using the same mothur-based sequence curation pipeline. After conducting an exhaustive review of the code they provided in their GitHub repository, I discovered several problems in the analysis that Martino and colleagues performed with the case study data. The details of this review can be found in the **Supplementary Text**. Solutions to the problems I identified are described below.

PERMANOVA analysis of case study datasets

- RPCA was compared to Bray-Curtis and weighted UniFrac distances Analysis compared F statistics, which again are flawed, since so much noise was removed
  - Sleep Apnea data analysis used three axes rather than two like the Sponge dataset
- RPCA not able to detect differences in CRC dataset
  - For BC and wU, euclidean distances were calculated from first two PCoA axes and used as input to PERMANOVA
  - when using RPCA all of the sequences in each sample were used but only 1,000 sequences per sample were used with the Bray-Curtis and weighted UniFrac distances (**Supplementary Text**).

KNN classification of case study datasets

- Used 1st 2 axes of PCoA of Bray-Curtis distances to calculate euclidean distances, which were then fed into kNN. Why not use the original Bray-Curtis distance matrix?
- I rewrote knn algorithm to use any distance matrix and to use 100 x 80:20 splits with search

for optimal k value

- Bray-Curtis was similar to better than RPCA for each dataset; other compositional methods performed at least as well as RPCA and better than Bray-Curtis for Sleep Apnea dataset

Sampling depth experiment with simulated and experimental data

- Buried in M&M was the description of an analysis and its results where Martino et al. compared the clustering of samples in an ordination where samples were duplicated and subsampled to 500 and 100 sequences per sample. They calculated distances using Jaccard, Bray-Curtis, and RPCA. They found that “RPCA alleviated the clustering by sequencing depth both qualitatively and quantitatively”
  - The ordinations still showed clustering by sampling depth
  - The Jaccard and Bray-Curtis distances were not calculated using rarefaction.
- Repeated analysis withe the simulated and experimental data by using 5000 and 10000 sequences per sample
- Analyzed distances usign PERMANOVA accounting for sampling depth -> Rarefaction was only method that could reliably control for the uneven sampling effort

Discussion

- Overall method problems
  - Choice of number of partitions/axes seems arbitrary/no guidance. Controls and sponges used 2, sleep apnea used 3. Default is 3
  - auto_rpca available within recent implementation but has since been deprecated
  - Authors acknowledged that RPCA doesn’t work on time series datasets
- Benchmarking should use methods that are commonly used in the field. KNN was interesting, but not relevant. Who is using raw with matrix completion?
- Statistical modeling approach is appealing for building communities and treatment groups and nicely complements results found using simulations based on observed sequencing

counts and sequencing depths

- Comment on problematic coding practices
- Use rarefaction

Materials & Methods

- Deicode benchmarking data and code
- KNN analysis
- Description of Sampling depth experiment
  - adonis2 model
- Workflow
  - Snakemake
  - Conda - Software versions
  - GitHub

Acknowledgements

- Thanks for making code publicly accessible!
- Funding agencies

References

1. **Schloss PD**. 2024. Waste not, want not: Revisiting the analysis that called into question the practice of rarefaction. mSphere **9**. doi:10.1128/msphere.00355-23.

2. **Schloss PD**. 2024. Rarefaction is currently the best approach to control for uneven sequencing effort in amplicon sequence analyses. mSphere **9**. doi:10.1128/msphere.00354-23.

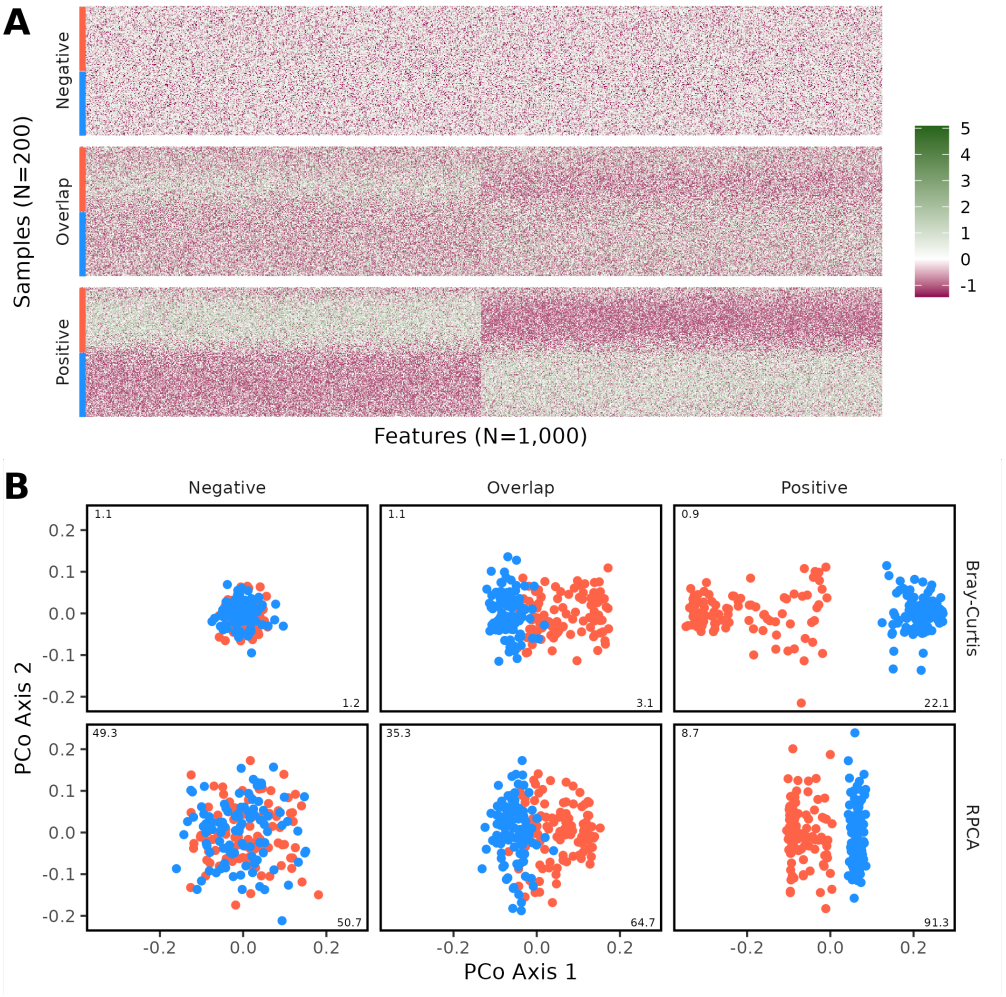


Figure 1. Graphical representation of an example dataset comparing communities simulated using code adapted from Martino when 4,000 sequences were sampled from each sample. This figure is analogous to Figure 3D of Martino with the inclusion of additional data from simulations where samples were shared between the two treatment groups and PCoA diagrams using Bray-Curtis distances in addition to RPCA. Heatmaps display the OCLR transformation of sequence counts. The numbers on the x and y axis of each ordination diagram represent the percent of the total variation that each axis explained.

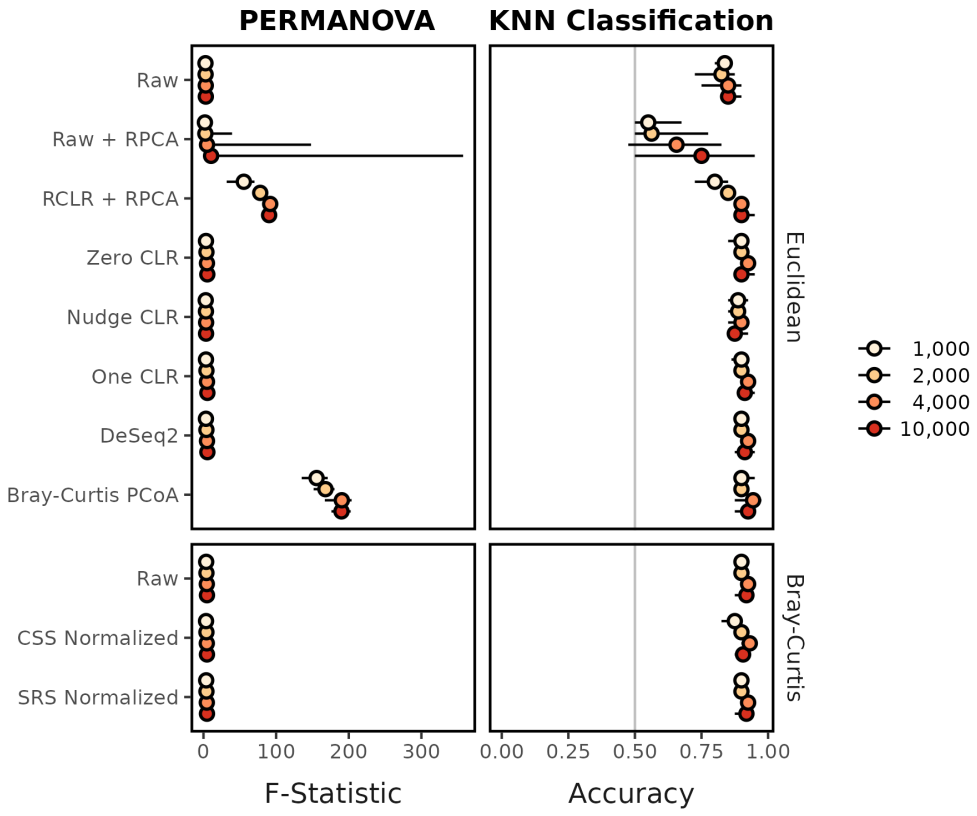


Figure 2. Comparison of RPCA's performance relative to other methods of comparing communities when using PERMANOVA and KNN classification to distinguish between overlapping treatment groups. This figure is analogous to Figure 3BC of Martino with the inclusion of results when using additional data transformations, Bray-Curtis distances, and 10 instead of 1 replicate simulation. Each circle represents the median of ten replicate simulations and the confidence line indicates the range between the minimum and maximum values. The median KNN classification accuracy and range were the median of 10 median values each calculated across 100 training-testing splits using the fitted K value from 80-20 splits. The vertical gray line for KNN classification indicates the accuracy one would expect from randomly guessing the treatment group.

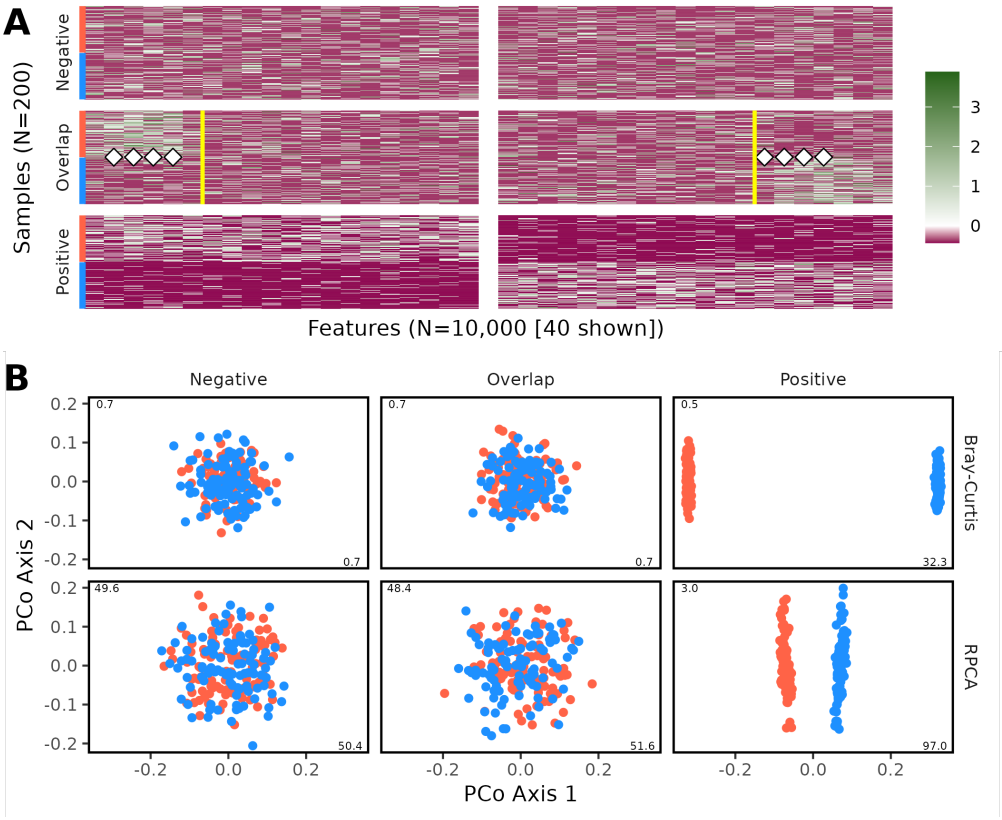


Figure 3. Graphical representation of an example dataset comparing communities simulated rewritten in R based on the description in Martino when 10,000 sequences were sampled from each sample. This figure is analogous to Figure 1. To highlight differences in overlap simulation, only the first and last 20 features from each sample are shown. The vertical white band indicates the break in the data presentation. The yellow vertical lines on the heatmap indicate the 12 features that are differentially represented in the two treatment groups. The white diamonds indicate features that had significantly different relative abundances when tested by a Wilcoxon test with p-values corrected using a Benjamini-Hochberg correction. For this example, the differences between the positive control (P=0.001) and overlap dataset (P=0.018) were significant by PERMANOVA and non-significant for the negative control (0.50) using Bray-Curtis distances and significant for the positive control (0.001) and non-significant for the overlap dataset (0.11) and negative (P=0.49) control using RPCA.

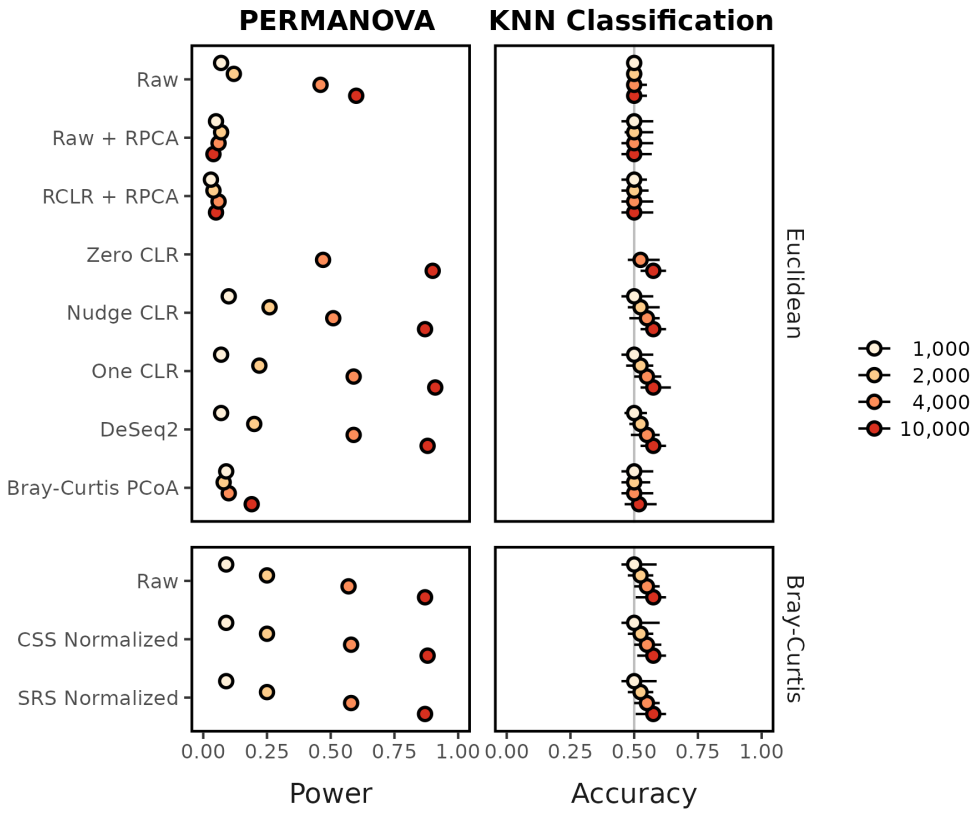


Figure 4. Comparison of RPCA's performance relative to other methods of comparing communities when using PERMANOVA and KNN classification with revised simulation approach to distinguish between similar treatment groups. This figure is analogous to Figure 2 except that it uses 100 instead of 10 replicates and the power to detect differences by PERMANOVA is shown instead of the Fst statistic. Each circle represents the median of the 100 replicate simulations and the confidence line indicates the range between the minimum and maximum values. The median KNN classification accuracy and range were the median of 10 median values each calculated across 100 training-testing splits using the fitted K value from 80-20 splits. The vertical gray line for KNN classification indicates the accuracy one would expect from randomly guessing the treatment group. The power and accuracy values for 1,000 and 2,000 sequences per sample when using Zero CLR were not calculated since the data were too sparse for the imputation algorithm to complete.

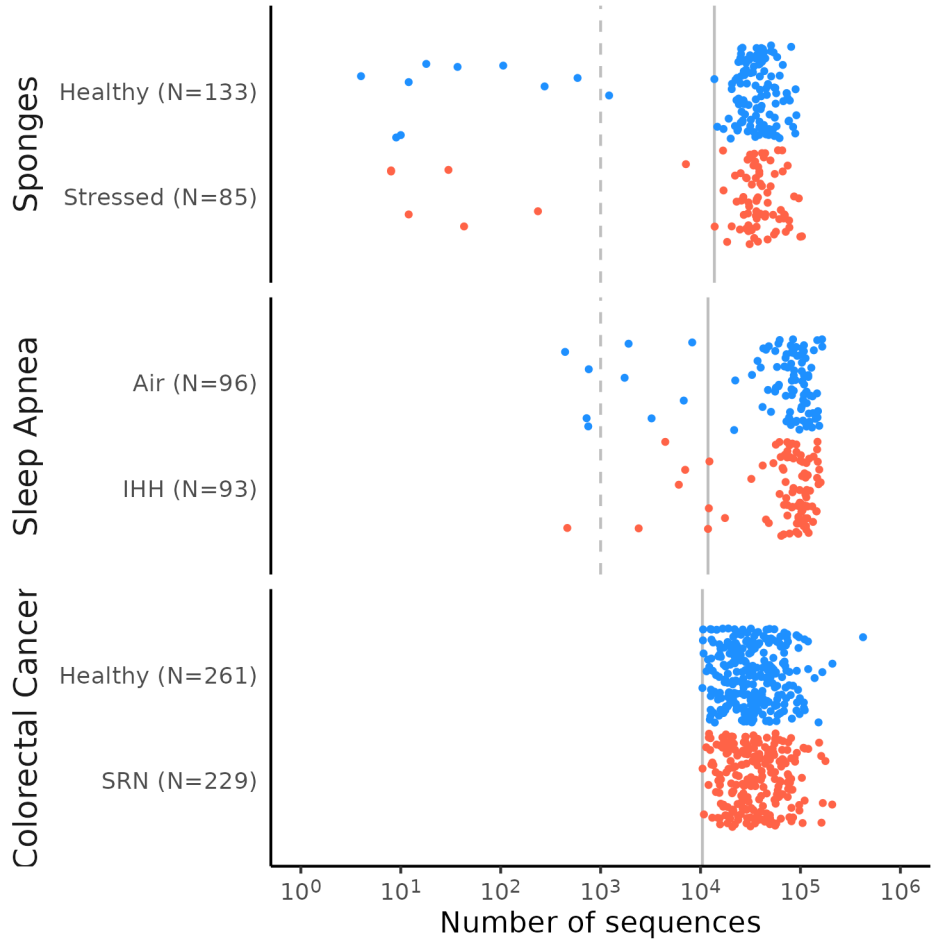


Figure 5. Comparison of the number of samples and variation in sequencing depth used in the three case studies. The vertical dashed line indicates the 1,000 sequence threshold, which was used by Martino and colleagues when calculating Bray-Curtis and weighted UniFrac distances. The vertical solid line indicates the higher threshold that was added in this study to explore the effects of sequencing depths. The higher threshold was 13755 sequences for the sponges dataset, 11,879 sequences for the sleep apnea dataset, and 10,439 sequences for the colorectal cancer dataset.

Supplemental Figures

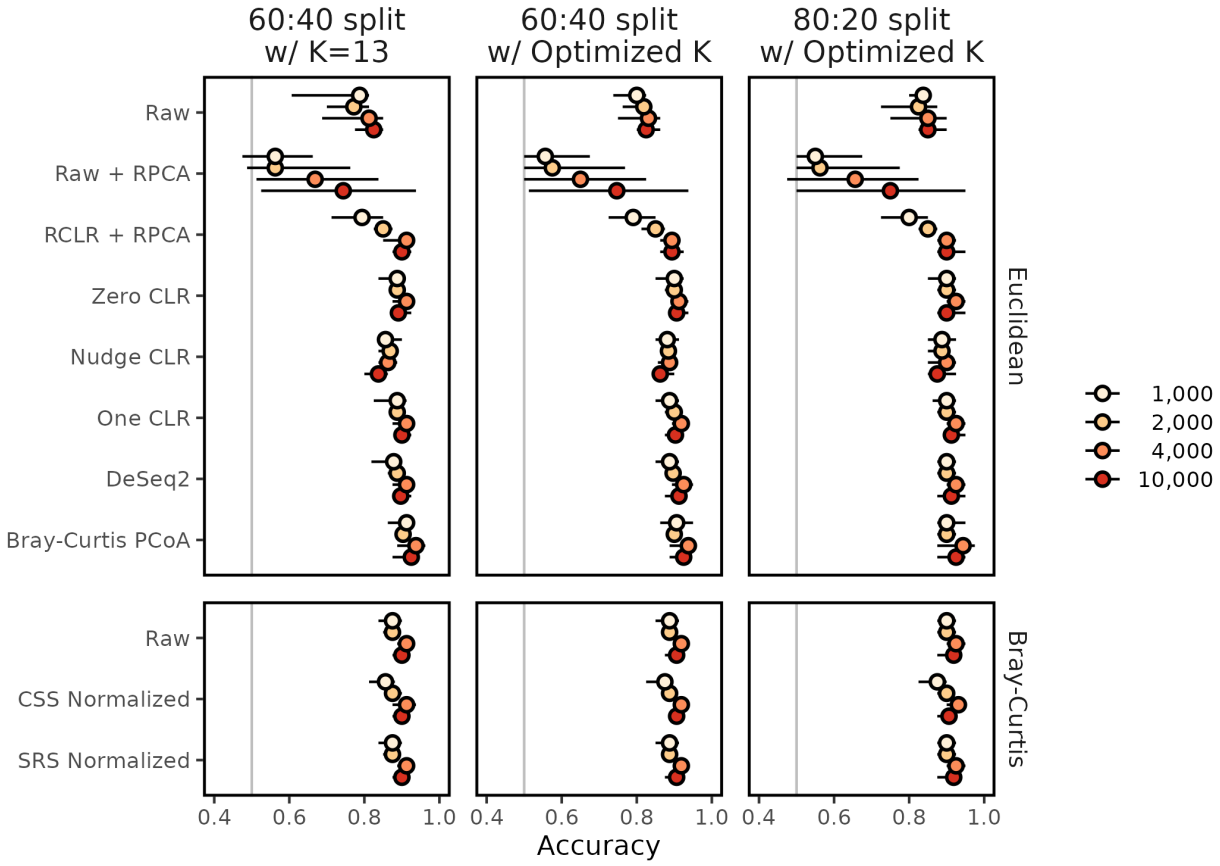


Figure S1. Differences in KNN classification accuracy for distinguishing between overlapping treatment groups using different approaches to optimize the number of neighbors was negligible. Martino split 60% of their data into a training set using 13 neighbors and tested on the remaining 40%. Following Topcuoglu et al. [**XXXXX**], I optimized the number of neighbors using 60-40 and 80-20 training-testing splits.

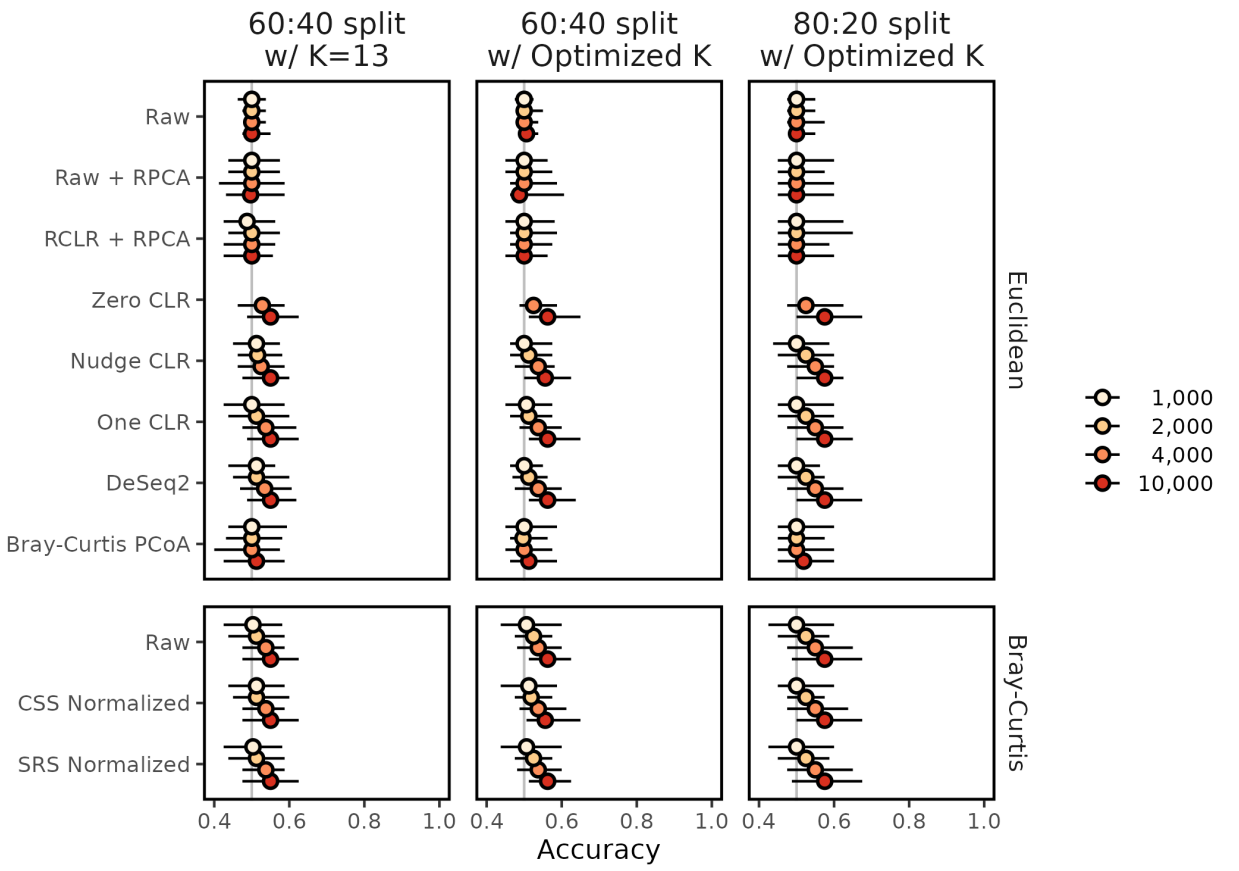


Figure S1. Differences in KNN classification accuracy for distinguishing between overlapping treatment groups using the revised simulation data by different approaches to optimize the number of neighbors was negligible. Martino split 60% of their data into a training set using 13 neighbors and tested on the remaining 40%. Following Topcuoglu et al. [**XXXXX**], I optimized the number of neighbors using 60-40 and 80-20 training-testing splits.